

Abdominal Pain in Second and Third Trimester

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Introduction

- Abdominal pain is one of the most common complaints in late pregnancy
- Pain can originate from pregnancy-related, gynecological, gastrointestinal, urinary, or musculoskeletal systems
- Identifying the cause is challenging because pregnancy changes can mimic disease
- Important to differentiate between benign/physiological and life-threatening causes

Abdominal Pain In Pregnancy

Abdominal pain in pregnancy is often due to a benign cause ,while there are many harmless causes but there may be serious cause.

Abdominal pain in pregnancy may be a normal part of the process as body changes as to accommodate growing baby.

Physiology

- Pain is a protective mechanism. It occurs whenever tissues are stretched, compressed, inflamed, or damaged. These changes activate nociceptors, which are pain receptors found in the uterus, ligaments, peritoneum, and abdominal organs.
- Three types of stimuli cause pain: mechanical, such as stretching of the round ligaments; thermal; and chemical, such as prostaglandins and inflammatory mediators.
- During pregnancy, mechanical stretching and chemical inflammation are the most common reasons for abdominal pain.

Two types of nerve fibers carry pain:

- A-delta fibers (Fast pain)
 - Myelinated → fast conduction
 - Sharp, localized pain
 - Example: sudden round ligament stretching
- C fibers (Slow pain)
 - Un myelinated → slow
 - Dull, diffuse, aching pain
 - Example: uterine stretching or GI discomfort
- Pregnancy abdominal pain is mostly C-fiber mediated → dull, gradual onset, poorly localized.

Physiology of Pain

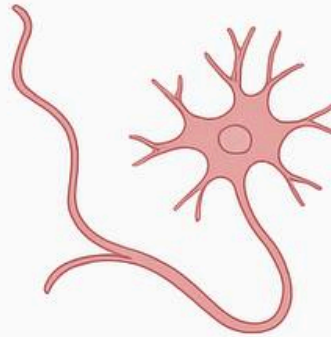
Pain is an unpleasant sensory and emotional experience produced when the body detects actual or potential tissue damage

- **Protective** – remove is harmful stimulus
- **Diagnostic** – remove or react to n andluy



Pain receptors

- Stretching
- Pressure



Nociceptors

the pain receptors

- **Specialized free nerve endings**
- **Since reacting**

Pregnancy sore
on the back

- **Pregnancy**

Is a condition

Three types of pain receptors

- **Mechanical**
 - Stretching, pressure, distension, compression, tearing or pulling
- **Chemical**
 - Prostaglandin synthesis and release
 - Adverse reactions
- **Pregnancy**
 - Uterine suprapubic pain and tenderness
 - Appendicitis, RLQ pain and
 - Chorioamnionitis, uterine tenderness

Types of pain fibers

A-delta fibers (Fast pain)

- Myelinated, myelinated, fast conduction
- Sharp, localized

C fibers (Slow pain)

- Dull, gradual onset, poorly localized

Obstetric Causes

We classified the pain to physiological and pathological causes

1) Physiological: benign, mild, intermittent, self-limiting pain not a sign of serious complication; related to anatomical and hormonal changes during pregnancy

- 1 Round ligament pain
- 2 Uterine stretching
- 3 Braxton hicks contraction
- 4 Digestive changes
- 5 Pelvic pressure ligament stretch

2) Pathological cause :Pathological pain indicates a potentially serious problem and requires evaluation we can subdivide to (obstetrical and Non obstetrical cause)
According to obstetrical cause they are pregnancy related

1- Preterm labor (most common in second trimester) (regular uterine contraction cause cervical dilation before 37week cause back pain then radiate to lower abdomen in Supra pubic region .

2 - Placental abruption (common in both second and third trimester) (characteristic presentation of placental abruption is that of painful bleeding associated with a tense rigid abdomen and uterus hard like a wooden board

3 - Uterine rupture (most common in third trimester) (usually occur in scared uterus previous c-section it cause sudden generalized abdominal pain due to tear wall thickness of the uterine wall

4 - Pre eclampsia (Hypertension after 20 weeks + organ dysfunction.

Mechanism of pain:Liver capsule stretching from hepatic edema.

5 - HELLP syndrome (is an acronym for hemolysis I elevated liver enzyme and low platelet present with epigastric pain nausea and vomiting associated with serious complication like acute renal failure placental abruption still birth)

6 - Acute fatty liver in pregnancy (cause epigastric pain and right upper quadrant pain due to liver swelling and metabolic failure)

7 - Acute polyhydramnios (most often occurs in second trimester present with rapid distension of the uterus, causing maternal pain and breathlessness. Mainly Upper Abdomen (Epigastric & RUQ/LUQ)

8 - Chorioamnionitis intra-amniotic infection usually due to bacterial infection and cause Lower abdomen pain and generalized uterine tenderness

9 - Pyelonephritis Kidney infection typically due to bacteria infection and cause Flank pain (R or L), may radiate to RUQ/back/lower abdomen

ectopic



Threatened
miscarriage

Pelvic
inflammatory
disease

Cholecystitis

~~ECTOPIC~~

Ovarian
cyst
rupture

Ovarian
torsion

Intestinal / GIT:

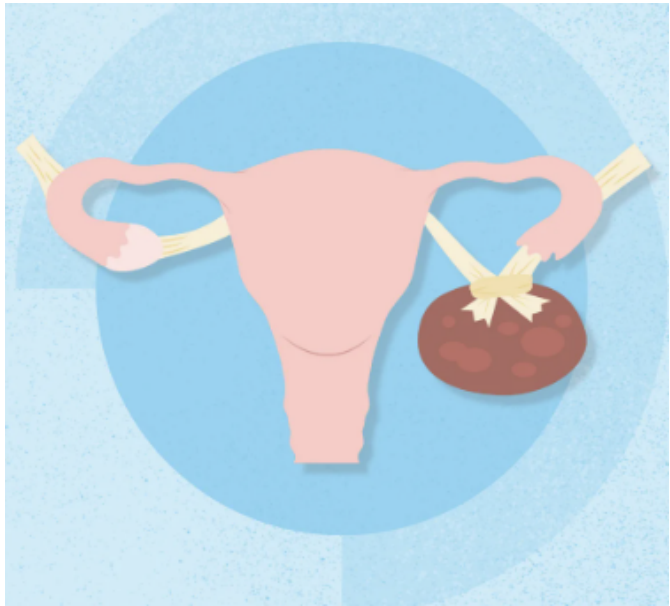
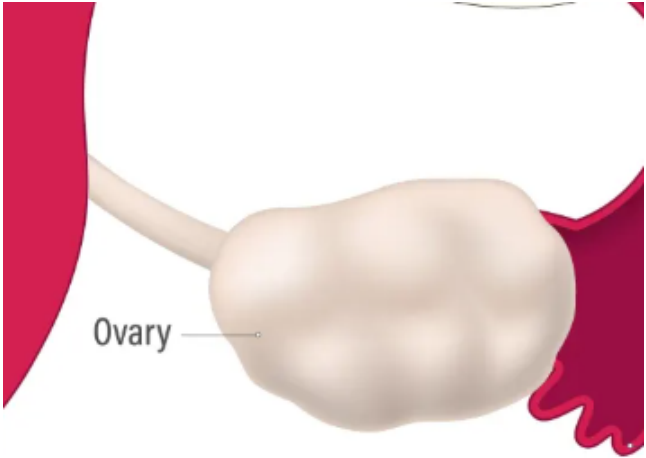
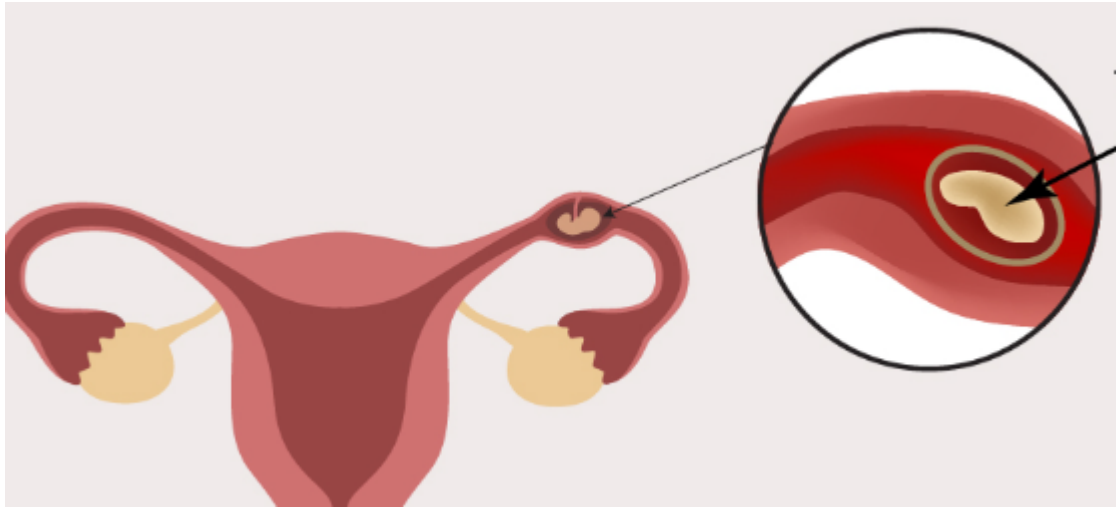
Appendicitis
Pancreatitis
Bowel
obstruction
Hernia

Urinary

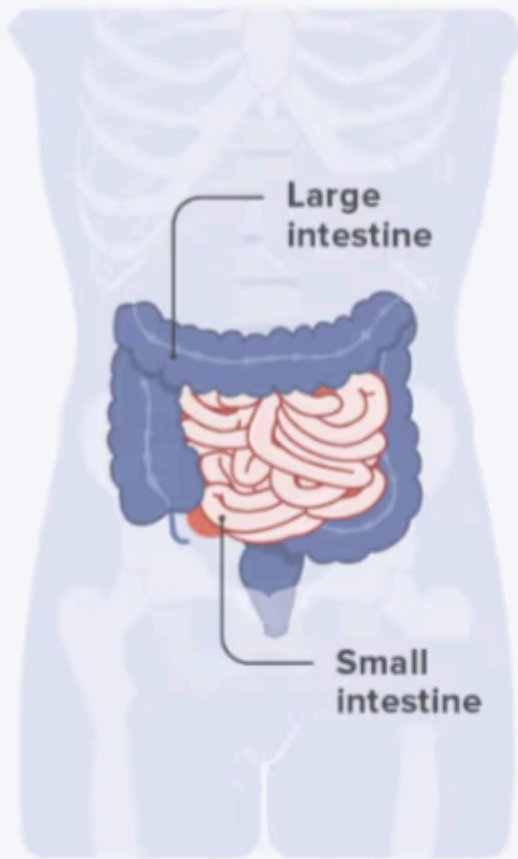
- Renal colic
- UTI

Fibroid degeneration

Muscle strain



Bowel Obstructions



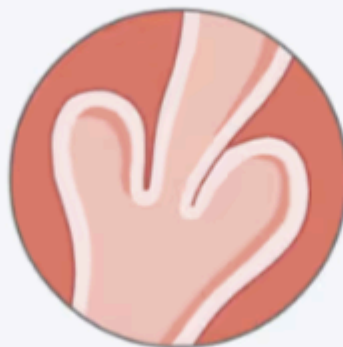
Herniation



Adhesions



Tumor/Cancer



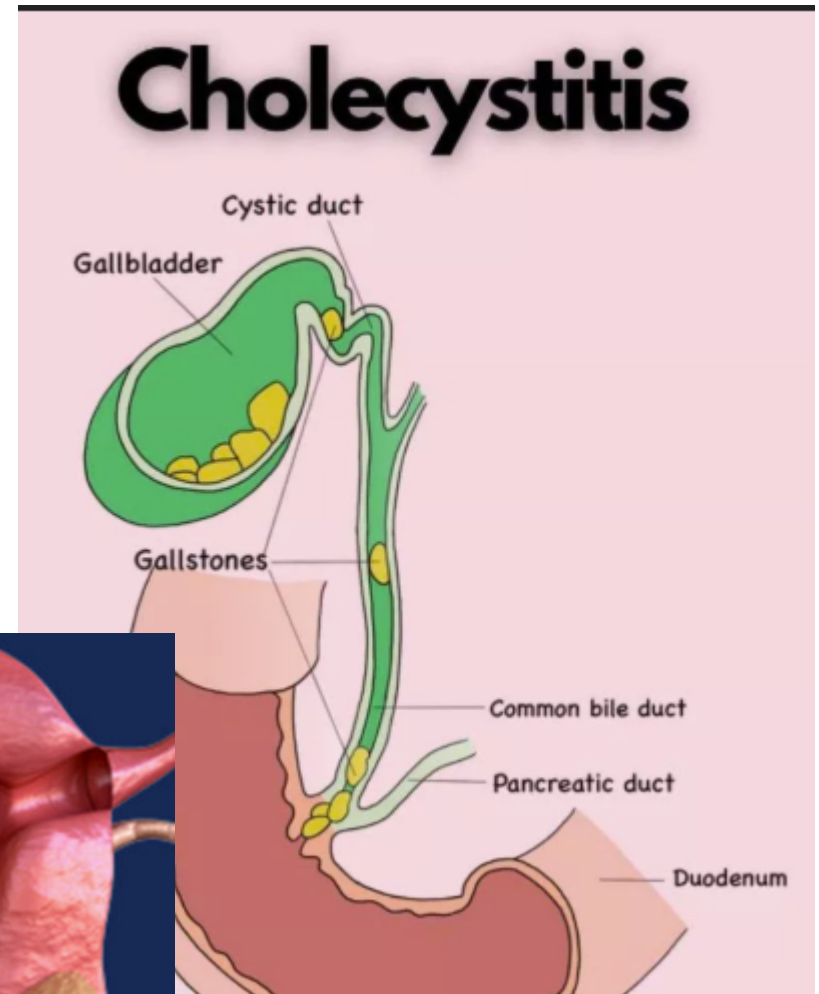
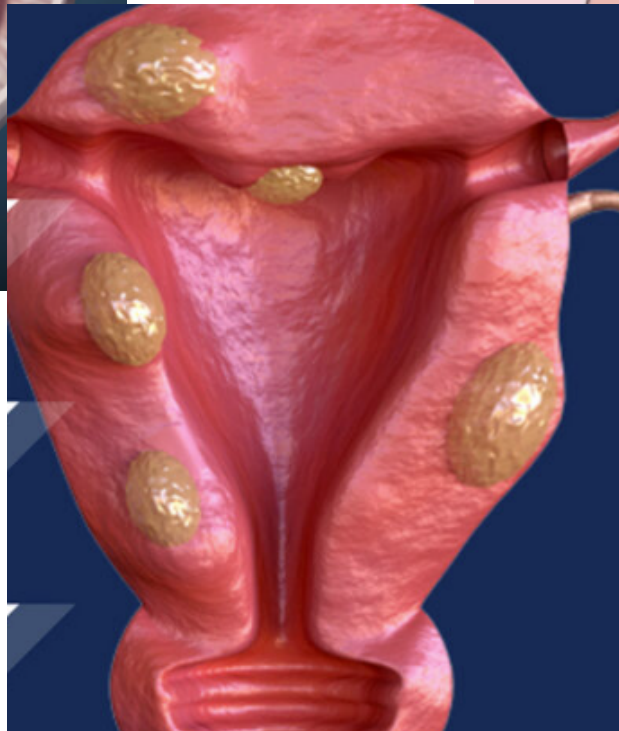
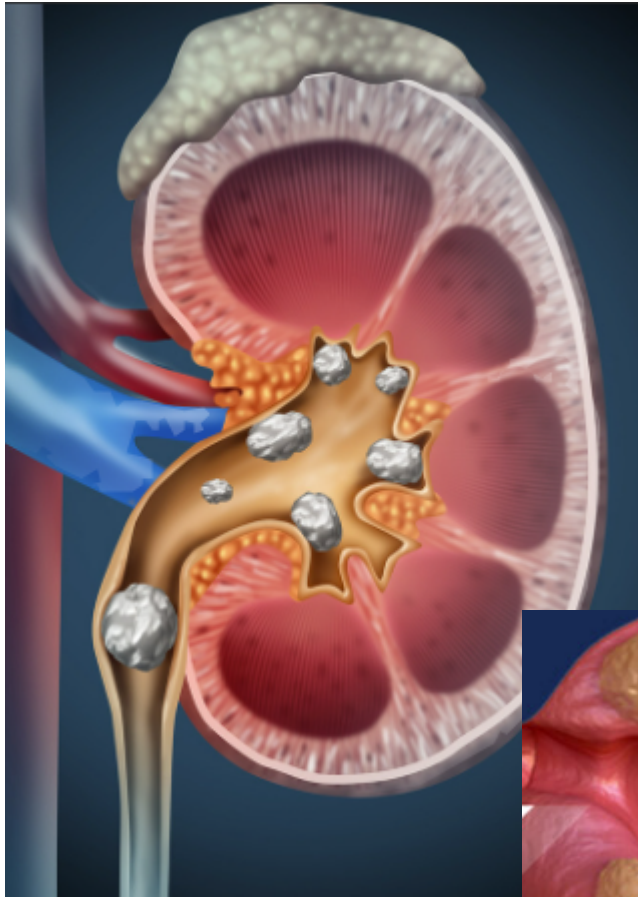
Intussusception



Volvulus



Foreign Object



4. Assessment & Investigations

- History:
 - Location, onset, severity, associated symptoms
 - Obstetric history, fetal movement
- Examination:
 - Vital signs, abdominal exam, uterine tenderness
 - Speculum exam if needed
- Investigations:
 - CBC, LFTs, renal tests
 - Urinalysis, CRP
 - Ultrasound (obstetric + abdominal)
 - CT/MRI only when essential

Common Non-Serious Causes (usually diagnosed clinically)

- Round Ligament Pain: A sharp, stabbing or aching pain in the lower abdomen or groin, often worse with sudden movement. Diagnosis is usually clinical after a physical exam and ruling out serious conditions.
- Constipation and Gas: Common due to hormonal changes and uterine compression. History and physical exam usually suffice; dietary advice (increased fiber, fluids) is the primary management.
- Braxton Hicks Contractions: Irregular, non-painful uterine tightenings that resolve with rest or rehydration. Fetal heart monitoring and confirmation that the cervix is not dilating help differentiate them from preterm labor.

3-investigations:

Initial investigations should be guided by clinical suspicion and aim to prioritize maternal and fetal safety while minimizing fetal exposure to radiation.

- **Laboratory Tests:**

- **Blood tests:** Complete blood count (CBC), liver and renal function tests, amylase/lipase (for pancreatitis), and a clotting screen if placental abruption is suspected.

- **Urine tests:** Urinalysis and culture to check for urinary tract infections (UTs) or pre-eclampsia (proteinuria).

Imaging:

- **Ultrasound (US):** The first-line imaging modality for most cases of abdominal pain in pregnancy as it is safe and can assess fetal well-being, the placenta, adnexal masses (ovarian cysts, torsion), gallstones, and hydronephrosis.

- **Magnetic Resonance Imaging (MRI):** A safe and reliable option for both mother and fetus when ultrasound findings are inconclusive, especially for conditions like appendicitis or intestinal obstruction.

MRI does not use ionizing radiation.

- **Computed Tomography (CT) Scan:** Generally avoided due to radiation exposure, but may be used in selected, urgent cases if the diagnosis cannot be made by other means and the benefit outweighs the risk.

Condition-Specific Management

Obstetric Causes

- Threatened miscarriage: observation, bed rest not required; follow-up ultrasound.
 - Ectopic pregnancy: stabilization; methotrexate or surgical management depending on stability and size.
 - Preterm labor: tocolytics, corticosteroids for fetal lung maturity, magnesium sulfate (neuroprotection), monitoring.
 - Placental abruption: stabilization, continuous fetal monitoring; urgent delivery if severe.
 - Round ligament pain: reassurance, heat therapy, gentle stretching.

Non-Obstetric Causes

- Urinary tract infection / pyelonephritis: pregnancy-safe antibiotics (e.g., cephalexin, nitrofurantoin—avoid near term).
 - Appendicitis: surgical consultation → appendectomy.
 - Cholecystitis / gallstones: hydration, analgesia; consider cholecystectomy if recurrent.
 - Renal colic: hydration, analgesia; ultrasound monitoring.
 - Gastroenteritis: hydration, antiemetics safe in pregnancy.

Supportive Treatment

- Adequate hydration.
 - Safe analgesics: paracetamol (first choice); avoid NSAIDs especially after 20+ weeks.
 - Antiemetics safe in pregnancy (e.g., doxylamine–pyridoxine, ondansetron with caution).
 - Monitoring: fetal heart rate, maternal vital signs, symptom progression.

Conclusion

Abdominal pain in pregnancy requires a careful, systematic approach because both obstetric and non-obstetric conditions may present similarly, and delays in diagnosis can pose risks to the mother and fetus. Early assessment, appropriate investigations (especially ultrasound), and condition-specific management allow for timely treatment. Most causes are benign, but serious conditions such as ectopic pregnancy, placental abruption, or appendicitis must be ruled out promptly. Effective management prioritizes maternal stabilization, fetal wellbeing, and individualized treatment based on the underlying cause.

References

- 1. Gynaecology by Ten Teachers
- Book
- 2 Clinical Obstetrics and Gynecology, 9th Edition – Speroff & Fritz
- 3 <https://www.ncbi.nlm.nih.gov/books/NBK499970/>